2022 International Conference on Machine Learning and Knowledge Engineering (MLKE) | 978-1-6654-9567-7/22/$31.00 ©2022 IEEE | DOI: 10.1109/MLKE55170.2022.00018

Behavioral biometrics dentification based on machine learning methods

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***Abstract*—This essay aims to develop a machine learning model that is able to build an initial advantage for behavioral biometrics. While behavioral biometrics have theoretically shown to be advantageous over traditional identification methods like passwords and physical tokens, more research needs to be conducted to prove that behavioral biometrics can ensure high accuracy in order to be applied in real life situations. This essay conducts an experiment that builds and compares three models that can potentially be used in this field. We use the data posted on the UCI Machine Learning Repository, which contains a set of motion data collected by phone accelerometers. This experiment has proved the feasibility of behavioral biometrics and set up the foundation for subject-wise studies.**

Keywords- Behavioral Biometrics; identification; machine learning; phone accelerometer

I. INTRODUCTION

Identification has been a highly-valued research topic, as it involves every aspect of human life, from national security to unlocking personal accounts. At present, the main methods of identification are passwords and physical tokens like fingerprints and iris. However, with the development of artificial intelligence and machine learning, behavioral biometrics emerge as an alternative way of identification, and some voices claim that it outweighs the traditional methods. Nevertheless, it is clear that passwords and physical tokens "can easily be stolen or duplicated" and therefore are not the best way to ensure security. Despite fingerprints and iris "can sometimes be difficult to use", [1] they do have concrete merits. First, the datasets can be easily obtained by scanning a fingertip or an iris. Second, this data-collecting process usually takes less than a minute. Third, the sizes of the data are usually small enough for normal calculation and storage. These advantages together make fingerprints and iris the most widely used security measures across the globe. Compared to them, behavioral biometrics have not gained many advantages, let alone that some people may not wear smartwatches or put their phones in their pockets. One feature of behavioral biometrics is still notable though – it can carry out the identification processes through subjects' movements without many environmental requirements, and therefore make the process more convenient. Comprehensively speaking, to make this identification method stand out among its peers, we need to simplify the data collected, shorten the training time, and improve the accuracy of the model. Therefore, this essay will touch on these topics by proposing a new model used for behavioral biometrics.

II. BACKGROUND AND RELATED WORK

Biometrics have long been a way of identifying biological individuals. Over the decades, this technology has been greatly refined in aspects of accuracy, convenience, and time complexity. Compared to knowledge-based authentication methods, biometrics shows "high efficiency, accuracy, and user acceptance." [2]

Much past research on behavioral biometrics was based on vision – datasets are processed in the form of image series, where movements can be recognized by silhouette separation or mathematical models.[3] However, this method has been deemed to be hard to execute. Nowadays, wearable sensor-based biometrics naturally draws researchers' attention. In this area, there were two main issues to be solved: first, the sensors were intrusive, since they are placed all over the subjects' bodies; and second, the accuracy of the model was still below expectation.[4] To solve these issues, it has long been proposed that "commercial smartphones and smartwatches" can serve as alternatives. The first related study was conducted by Mantyjarvi et al. in 2005, when various machine learning methods were not developed yet, and the accelerometers were not mature enough to make precise measurements. The accuracy of the method mentioned in the study was not much better than vision-based biometrics by then – the results floated between 72% and 88%.[5]

In 2019, Weiss et al. further improved smartphone-based biometrics by measuring a wider variety of activities, and thus build a model that fits into a more general setting. This study also includes datasets collected by sensors on smartwatches, which are getting accepted by the market these years. However, they will not be the focus of this essay since they have not been prevalently used by people at present.

This essay will use the datasets collected by phone accelerometers in Weiss et al.'s study (which are posted on UCI Machine Learning Repository) and devise a new model for identifying subjects. Decision Trees, SVM, and Deep Neural Networks are deployed, compared, and refined in this experiment.

III. METHODOLOGY

This study implemented three algorithms: Decision Trees, SVM, and DNN.

A. Decision tree

Decision Trees are a supervised machine learning method. It sets out with a given set of data, as well as some if-else decision rules. Then, it will learn data features based on the

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DOI 10.1109/MLKE55170.2022.00018

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rules and build an approximation model, a tree, to fit the dataset. The tree takes the form of decision nodes and leaf nodes. Each decision node has two or more branches that represent different situations and properties; each leaf node represents the outcome – that is, a classification or a decision. Target values can therefore be predicted using the model. The complexity of decision rules depends on the depth of trees. The deeper the trees, the more accurate model we can get, and the higher capability the model has to make predictions. This algorithm is mainly utilized for classification and registration. In this experiment, we use Decision Trees because it is not sensitive to the size of a dataset – which means that it has advantages in adapting to datasets of different sizes.

B. SVM

Support-Vector Machines (SVM) are supervised machine learning methods also widely used for classification and regression. The mechanism of this algorithm is to first map the data in space, and then calculate a hyperplane that can a) divide the data into two categories, and b) maximize the distances between the function and the nearest points on both sides. The central point is to create the clearest gap between the two categories to maximize the prediction accuracy of the model. The phrase support vector refers to the vectors that define the hyperplane. When predicting target values, SVM maps them in the same space and decides their categories based on which side of the hyperplane they are in. To calculate the SVM classifier is to minimize the following expression:

$$\left[\frac{1}{n}\sum_{i=1}^{n}\max\left(0, 1 - y_i(\mathbf{w}^T\mathbf{x}_i - b)\right)\right] + \lambda\|\mathbf{w}\|^2.$$

This can be made more straightforward by looking at the graph below:

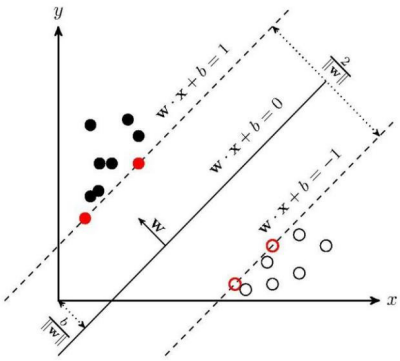


Figure 1. SVM Linear classification

From the graph, we can easily calculate the linear function which can best cut out a gap between the two clusters. However, to cope with non-linear classification situations, we need to apply a kernel trick to the hyperplane. We replace every dot product in our calculation with a non-linear kernel function, which helps to find a best-fit hyperplane within a transformed, higher-dimensional feature space. This hyperplane serves as the optimal classifier for the database.

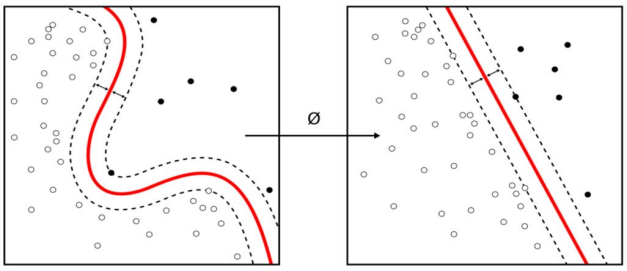


Figure 2. SVM Non-linear Classification

SVM is used in the experiment because it has strength in classifying datasets of relatively small size (we have truncated the original dataset to facilitate the experiment).

C. DNN

Deep Neural Networks (DNN) are, as the name shows, inspired by structures in human brains. Other than neurons and synapses, DNN also has weights, biases, and functions. To train a model that fits the datasets, DNN first creates a map that contains a group of interconnected neurons, which are shown in the figure below:

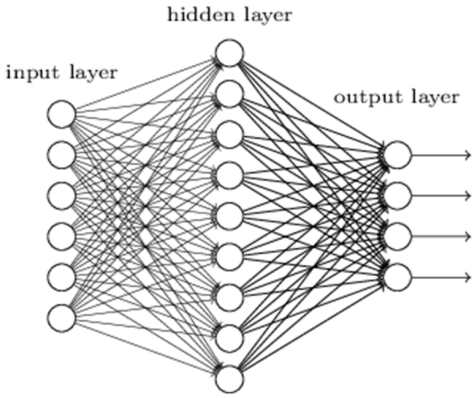


Figure 3. DNN Map

This map consists of multiple layers of neurons, including the input and output layers. Then, a random value is assigned to each connection as weight. The calculation processes start here. DNN multiplies input values by the assigned weight, and the data flows towards the output layer. There, an output between 0 and 1 is released. DNN would self-adjust the weight of neuron connections if the result does not successfully recognize the pattern of the datasets. Compared to the former two methods, DNN is especially good at calculating non-linear models that can better fit the datasets. Furthermore, it has advantages in directly extracting patterns from the data and building functions, as well as in high generalization capabilities. These highlights can be seen from the experiment results shown in the next section.

IV. Experiments and Results

In this experiment, we use the WISDM datasets posted on UCI Machine Learning Repository. These datasets include 18 distinct activities performed by 51 subjects and recorded by phone accelerometer, phone gyroscope, watch accelerometer, and watch gyroscope respectively. The data takes the form of

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subject labels, timestamps, activity labels, and three-dimensional coordinates recorded at a given time.

TABLE I. DATASET SAMPLE INFORMATION

ACTIVITY	X0	X1	X2	…	RESULTANT	Class
A	0.235	0.47	0.275	…	10.0518	1600
A	0.275	0.44	0.27	…	10.1171	1600
B	0.365	0.53	0.095	…	10.2354	1600
B	0.3	0.425	0.17	…	10.9159	1600

In this essay, we only use data collected by phone accelerometers, since this is the most well-developed technology at present. We also shrunken the dataset from 51 subjects to 44 subjects, while still keeping it large enough to support the experiment.

In this experiment, we would like to explore the effect of PCA on the accuracy and time complexity of the model. We would also like to compare the effectiveness of the three algorithms mentioned above in doing behavioral biometrical identification. Last, we would like to test if joining the three algorithms and adding a voting technique will improve the accuracy.

Principle Components Analysis (PCA) is a well-known method used to reduce the dimensionality of a large dataset. The general principle of this method is to extract the main information from the data, so that only part of the original dataset needs to be analyzed. This can sometimes significantly reduce the time complexity of a model since PCA essentially shrinks the dataset into a smaller one and excludes extraneous variables that may misguide the result. However, it also leads to an inevitable loss in accuracy, as some important information might be filtered out during the process. Therefore, choosing a suitable dimension for PCA means finding a balance between time complexity and accuracy of a model.

In this part of the experiment, we would explore the trend of accuracies of the three algorithms under the influence of PCA dimensionality reductions. The accuracy results are shown in figure 1, with a maximum of 20 dimensions reduced, ranging from 44 to 25. Then, the tendencies of the accuracy results are also shown in table 2:

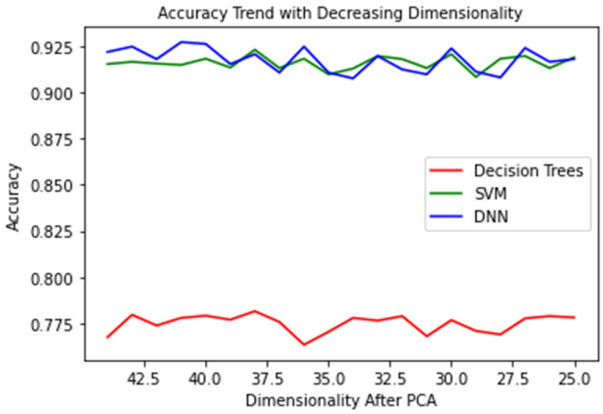


Figure 4. Accuracy results with PCA

TABLE II. ACCURACY RESULTS TENDENCY OF DIFFERENT METHODS

Dimension \ Algorithm	Decision Trees	SVM	DNN
44	0.768	0.915	0.921
43	0.780	0.916	0.925
42	0.774	0.915	0.918
41	0.778	0.915	0.927
40	0.779	0.918	0.926
39	0.777	0.913	0.915
38	0.782	0.923	0.921
37	0.776	0.913	0.911
36	0.763	0.918	0.925
35	0.770	0.910	0.911
34	0.778	0.913	0.908
33	0.777	0.920	0.920
32	0.779	0.918	0.912
31	0.768	0.913	0.910
30	0.777	0.921	0.924
29	0.771	0.908	0.911
28	0.769	0.918	0.908
27	0.778	0.920	0.924
26	0.779	0.913	0.916
25	0.778	0.919	0.918

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From table 2, we can see that although the overall accuracy is not heavily influenced by dimensional reduction, they fluctuate up and down after a dimension of 40. Therefore, to secure a stable output of accuracies, we will use PCA to reduce the dimension to 40 in later parts of the experiment.

The second part of the experiment concerns the effectiveness of the three methods introduced: Decision Trees, SVM, and DNN. Under the dimension of 40, the output of the three algorithms is plotted as columns shown below:

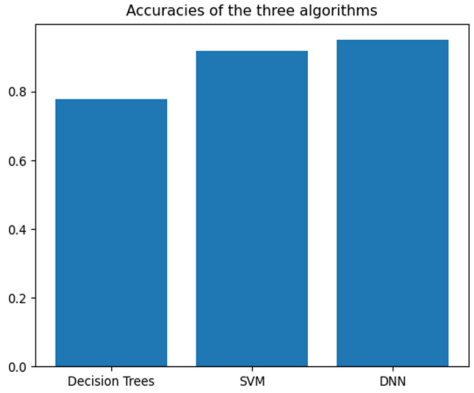


Figure 5. Accuracy results of different methods

TABLE III. COMPARISON RESULTS OF DIFFERENT METHODS

Algorithm	Decision Trees	SVM	DNN
Accuracy	77.9%	91.8%	95.1%

The disadvantage of Decision Trees is out of question. The biggest advantage of Decision Trees, that it is insensitive to the size of database and therefore highly adaptive, also leads to one of its main fallbacks – a relatively low accuracy. One reason for this might be that the model creates trees that are too complex – they fit the training set too well but prove to be bad at generalizing and applying it to the testing set. Another reason could be that Decision Trees are sensitive to small variations in the database, and are therefore unstable in output. This can be proved by the graph in the PCA experiment: compared to the other two methods, Decision Trees show greater fluctuation, as each reduction in dimension creates a large number of variations in the dataset.

SVM and DNN show greater capability in the experiment. Since we have shrunken the dataset, it creates a natural advantage for SVM and makes it comparable to DNN. As to the parameter of SVM, we use Redial Basis Function (RBF) as its kernel. Because RBF is able to map the sample to a higher-dimensional space, it can handle the situation where class labels have a non-linear relationship with their properties. Furthermore, it has fewer parameters than other kernels, which makes it less complex and therefore less numerical difficulties when doing calculations. As to the penalty parameter c, we set it to 1000, for we would like to build a model specifically used for behavioral biometrics. Since we do not require it to be highly adaptable to other application scenarios, we desire a high penalty coefficient to raise the accuracy of SVM. These advantages and parameter adjustments together help SVM to achieve a high accuracy almost equivalent to neural networks.

Neural networks have been proven to be one of the most popular and effective methods in machine learning. To build a DNN model, we use the MLPClassifer from the sklearn pack.

To choose the most suitable solver, we make a comparison among the three solvers – adam, lbfgs, and sgd. Here is a graph of accuracies of the model under different solvers (we keep other parameters constant to facilitate the contrast):

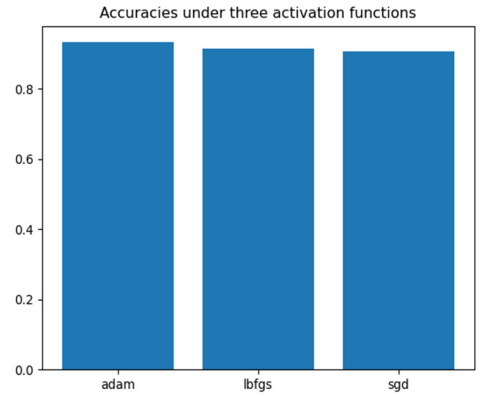


Figure 6. Accuracy with different solvers

TABLE IV. COMPARISON RESULTS OF DIFFERENT SOLVERS

Activation Function	adam	lbfgs	sgd
Accuracy	93.2%	91.5%	90.7%

We choose “adam” as the solver to optimize the weight of each connection, since “adam” is proven to be more effective when dealing with a relatively large sample (several thousand or more). Also, according to the experiment, adam has a significantly smaller number of iterations, which saves time without sacrificing accuracy.

As to the activation function, we do the experiment once again.

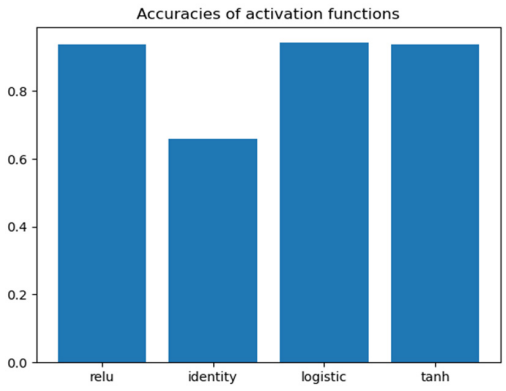


Figure 7. Accuracy with different activation functions

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TABLE V. COMPARISON RESULTS OF DIFFERENT ACTIVATION FUNCTIONS

Activation Function	relu	identity	logistic	tanh
Accuracy	93.8%	65.9%	94.2%	93.6%

We will use “logic” in the following experiment as it has been proven to be suitable for this scenario.

To further improve the accuracy, we raise the number of neurons in the hidden layer. Because a higher accuracy inevitably leads to a compromise for time complexity, we set the neurons to (400, 200). These parameter settings contribute to the overall accuracy of the DNN model and help it reach an accuracy of 95.06%.

From the experiment results, DNN has an unsurmountable advantage over Decision Trees and SVM. One of the biggest reasons behind this is that DNN is highly flexible and able to adapt itself to datasets with non-linear relationships. While neural networks have been proven to be highly efficient in doing pattern recognition in computer visions, it is now preliminarily tested to be also suitable for behavioral biometrics.

V.CONCLUSION

In this paper, a multi-machine learning method is proposed to conduct behavioral biometrics identification.According to the experiment results, the proposed method could achieve a satisfying result, which has shown a quite promising future. Since phone accelerometer has been prevalently used and has proven to be effective in behavior recognition, behavioral biometrics may appear as a better alternative to current identification methods.

This research has laid a foundation for further field-oriented study. We may examine whether behavioral biometrics can successfully be applied to real-life situations. For example, further research could focus on improving and specializing machine learning models in fields of medical science, national security, psychology, etc.

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